

Assignment 11

1. Calculate the stoichiometric oxidizer-fuel mass ratio (r_{stoic}) for the $\text{CH}_4 - \text{O}_2$ reactant mixture. {Ans.: 4}
2. Using the CEA software, calculate the combustion temperature (also known as the flame temperature) of a $\text{CH}_4 - \text{O}_2$ rocket thrust chamber with $r = 4$. The combustion chamber pressure is 68 atm. The reactants enter it at 298.15 K. Also list the species formed by the reaction. Ignore any species with mole fraction less than 0.01. {Partial Ans.: 3625 K}
3. Assuming monatomic, diatomic and triatomic species to have specific heat ratios of 1.67, 1.29 and 1.17, respectively, calculate the specific impulse of the rocket thrust chamber of Problem 2 in sea-level operation. The isentropic nozzle has area ratio 10. Is the exhaust jet under-, fully-, or over-expanded? {Ans.: 285 s (intermediate results: $R_m = 361.5 \text{ J/kg-K}$, $\gamma = 1.21$)}
4. A rocket thrust chamber has an isentropic nozzle with contraction ratio (A_c/A_t) of 2. Find the Mach number at the exit of the combustor. Also find the percent loss in stagnation pressure across it. $\gamma = 1.25$. {Partial Ans.: 5.23%}